

E#3 - Module 3: In-Class Exercises #3

Formal Definition of the FSA

A Finite State Automaton is defined as a 5-tuple:

$$M = (Q, \Sigma, \delta, q_0, F)$$

- **Q (States):**

$$Q = \{ q_0, q_1, q_5, q_{10}, q_{15}, q_{20}, q_{25}, q_{30} \}$$

(Complete set for pennies): $Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{14}, q_{15}, q_{16}, q_{17}, q_{18}, q_{19}, q_{20}, q_{21}, q_{22}, q_{23}, q_{24}, q_{25}, q_{26}, q_{27}, q_{28}, q_{29}, q_{30} \}$

(Complete set for pennies):

$$Q = \{ q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8, q_9, q_{10}, q_{11}, q_{12}, q_{13}, q_{14}, q_{15}, q_{16}, q_{17}, q_{18}, q_{19}, q_{20}, q_{21}, q_{22}, q_{23}, q_{24}, q_{25}, q_{26}, q_{27}, q_{28}, q_{29}, q_{30} \}$$

(Optional: add q_{dead} for overshoot handling)

All intermediate states for pennies (q_2, q_3, q_4 , etc.).

Each state represents the cumulative amount entered so far.

- **Σ (Alphabet):**

$$\Sigma = \{ 1, 5, 10, 25 \}$$

Where:

- 1 = penny
- 5 = nickel
- 10 = dime
- 25 = quarter

- **q_0 (Initial State):**

$$q_0 \text{ (represents 0¢)}$$

- **F (Final States):**

$$F = \{ q_{30} \}$$

(Note: $q_{\text{dead}} \notin F$)

- \$ (acknowledgement state when 30¢ is reached)

- **δ (Transition Function):**

For each state, add the coin value if the sum ≤ 30 ¢.

Examples:

- $\delta(q_0, 5) = q_5$
- $\delta(q_0, 10) = q_{10}$
- $\delta(q_5, 25) = q_{30}$
- $\delta(q_{10}, 10) = q_{20}$
- $\delta(q_{25}, 5) = q_{30}$

Full rule:

For all $i \in \{0..30\}$ and coin $c \in \Sigma = \{1,5,10,25\}$:

If $i + c \leq 30$: $\delta(q_i, c) = q_{i+c}$

If $i + c > 30$: $\delta(q_i, c) = q_{\text{dead}}$ (optional for DFA completeness)

For q_{30} (accepting sink): $\delta(q_{30}, c) = q_{30}$ for all $c \in \Sigma$

For q_{dead} (if used): $\delta(q_{\text{dead}}, c) = q_{\text{dead}}$

If adding a coin exceeds 30¢, ignore or stay in q_{30} .

Full rule:

For all $i \in \{0..30\}$ and coin $c \in \Sigma = \{1,5,10,25\}$:

If $i + c \leq 30$: $\delta(q_i, c) = q_{i+c}$

If $i + c > 30$: $\delta(q_i, c) = q_{\text{dead}}$ (optional for DFA completeness)

For q_{30} (accepting sink): $\delta(q_{30}, c) = q_{30}$ for all $c \in \Sigma$

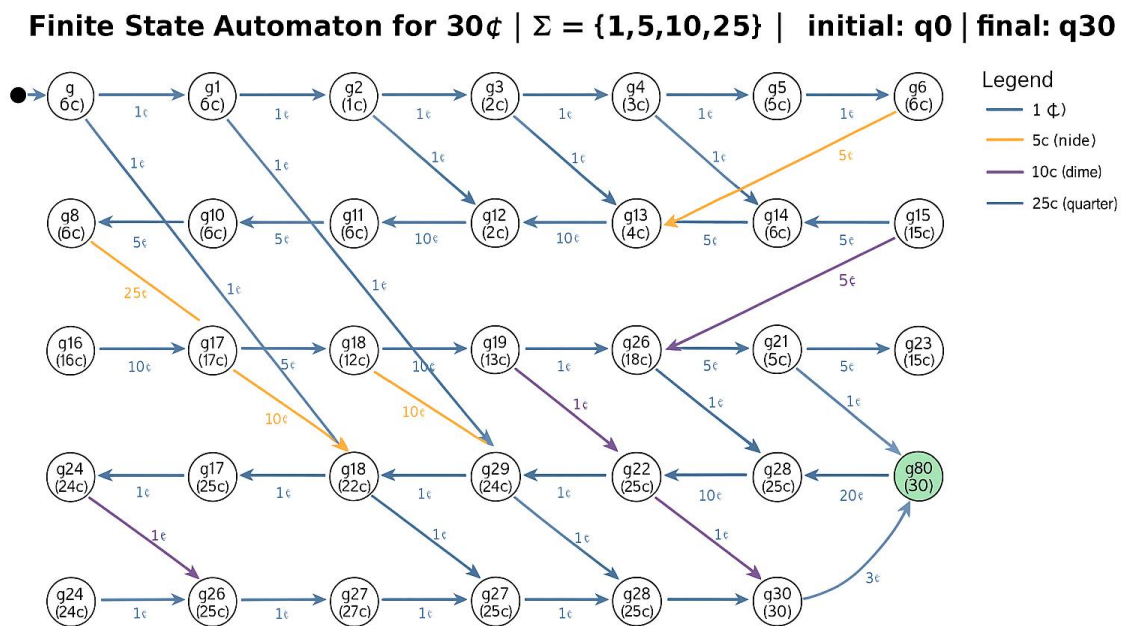
For q_{dead} (if used): $\delta(q_{\text{dead}}, c) = q_{\text{dead}}$

Complete Transition Function (δ)

Diagram Structure

- **Nodes:** $q_0 \rightarrow q_1 \rightarrow q_5 \rightarrow q_{10} \rightarrow q_{15} \rightarrow q_{20} \rightarrow q_{25} \rightarrow q_{30}$
- **Edges:** Labeled with coin values (1, 5, 10, 25)
- **Final State:** q_{30} (double circle)
- **Initial State:** q_0 (arrow pointing to it)

The diagram should include all states $q_0 \dots q_{30}$ and show q_{30} as a double circle with a self-loop for Σ .



Example Transition Table

Current State Input Next State

q_0 1 q_1

q_0 5 q_5

Current State Input Next State

q_0 10 q_{10}

q_0 25 q_{25}

q_5 5 q_{10}

q_5 10 q_{15}

q_5 25 q_{30}

...